Debiased Counterfactual Estimation for Nonlinear Panel Factor Models: A Generative Deep Learning Approach*

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Abstract

Panel methods such as generalized synthetic control use a linear factor model both to adjust for unobserved differences and to predict untreated outcomes. Predictions can fail when units respond nonlinearly to common conditions, even if the factors are estimated well. We keep factor-based adjustment and use generative adversarial networks for flexible prediction. For average effects on treated units, weighted control residuals correct prediction errors. For quantile effects, a separate regression and residual generator estimate the distribution of untreated outcomes. We establish identification and consistency and develop a framework for bootstrap inference. In simulations, the average-effect estimator is more accurate than augmented synthetic control, difference-in-differences, and synthetic difference-in-differences. When the covariate distribution shifts, its root mean squared error is 78 percent below generalized synthetic control's. The distributional estimator recovers quantile effects more accurately overall than group-level distributional methods. An application to political connections during the 2008 financial crisis finds that connected firms' returns exceed their estimated counterfactual most in the lower part of the distribution.

Keywords: panel data; causal inference; Wasserstein GAN; counterfactual estimation; double robustness; AIPW; nonlinear factor model

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1 Introduction

Causal panel studies must reconstruct the trajectory treated units would have followed without treatment. Latent factor models address this problem by representing unobserved unit types through loadings on common time factors. Pre-treatment outcomes help recover those types for adjustment and counterfactual prediction. Generalized synthetic control (GSC) and related estimators make this structure central to panel inference (Xu 2017; Ben-Michael et al. 2021; Arkhangelsky et al. 2021; Athey et al. 2021).

In linear factor methods such as GSC, the factor model does two jobs at once: it captures how units differ, and it fixes the form of the prediction. Units need not respond linearly, however. A thinly capitalized bank may barely react to mild funding stress but suffer large losses under severe stress, while a well-capitalized bank absorbs both. A linear predictor can get these responses wrong even when the factors themselves are estimated well.

Averages can also hide where an effect occurs. Political connections may protect firms from severe losses without changing their typical outcomes. To see this, we need the distribution of untreated outcomes, not just its mean. Comparing its quantiles with those of treated outcomes gives marginal quantile treatment effects (QTEs) (Firpo 2007; Gunsilius 2023).

Our approach separates the factor model's two jobs. We first estimate factors and loadings for adjustment. Neural networks then learn how untreated outcomes depend on them and on observed covariates, starting from a linear factor prediction, as in GSC. This lets the response be nonlinear without discarding the information in the panel's pre-treatment history.

Feng (2026) uses local PCA to recover confounder information from nonlinear proxies and develops orthogonal inference for mean and distributional effects. We take a different route: we assume a strong, recoverable linear factor component, estimate it first, and let neural networks learn the nonlinear response that it leaves out.

For ATT, DR-FGAN combines a Wasserstein adversarial mean learner with gradient penalty (Gulrajani et al. 2017), cross-fitting (Chernozhukov et al. 2018), and augmented inverse-probability weighting (AIPW) (Robins et al. 1994). The residual correction gives

the ATT estimator its double robustness (Bang and Robins 2005). For QTE, a separate regression estimates the untreated mean and an empirical-transport Cramér GAN (ETC-GAN) learns the conditional residual distribution. We add the mean back to these residuals and pool the resulting distributions over treated units. Their quantiles describe what the treated group's outcomes would have looked like without treatment.

We establish identification and consistency conditions for both estimators and study how their prediction errors affect the estimated treatment effects. Adversarial critics compare observed and generated outcomes together with their predictors (Goodfellow et al. 2014), so training can respond to distributional differences that a few fixed moments miss. For inference, we resample whole units and refit the estimator. Treating each unit's path as a sampling cluster allows its observations to be dependent and heteroskedastic. The Supplementary Information (SI) gives proofs and implementation details; panel diagnostics follow Liu et al. (2024).

The simulations show when nonlinear prediction pays off. With a stable covariate distribution, DR-FGAN estimates ATT more accurately than augmented synthetic control (ASC), difference-in-differences with two-way fixed effects (DID/TWFE), and synthetic difference-in-differences (SDID). When the distribution shifts, the nonlinear response no longer averages out, and DR-FGAN's root mean squared error (RMSE) is 78 percent below GSC's. ETC-GAN estimates quantile effects more accurately overall than group-level changes-in-changes (CiC) and CDF-DiD. In the political-connections application, connected firms' returns exceed their estimated counterfactual most in the lower part of the distribution. A supplementary electricity application examines household responses to dynamic tariffs and colder weather.

2 Setup and Identification

2.1 Data and Target

Consider a balanced panel of $N = N_{\text{tr}} + N_{\text{co}}$ units and T periods, with treated and control groups $\mathcal{T}$ and $\mathcal{C}$. Treatment begins at $T_0 + 1$; D_i denotes group membership and $D_{it} = D_i \mathbf{1}(t > T_0)$. We observe (Y_{it}, D_{it}, X_{it}), where X_{it} may include time-invariant characteristics.

The population targets are

$$ATT_t = \mathbb{E}\{Y_{it}(1) - Y_{it}(0) \mid D_i = 1\}, \qquad ATT = \frac{1}{T - T_0} \sum_{t > T_0} ATT_t. \tag{1}$$

In estimation, we replace these expectations with averages over the treated units. Since treated post-treatment $Y_{it}(1)$ is observed, estimation requires the missing $Y_{it}(0)$.

For $t > T_0$, the marginal quantile treatment effect is

$$QTE_t(\tau) = Q_\tau\{Y_t(1) \mid D = 1\} - Q_\tau\{Y_t(0) \mid D = 1\}, \qquad \tau \in (0, 1). \tag{2}$$

It compares the same percentile of the treated group's outcome distribution with and without treatment. Here, marginal refers to the distribution across the treated population, averaging over its different characteristics. QTE compares positions in these two distributions rather than treatment effects for particular units. ATT describes the average effect; QTE describes how the contrast varies across the distribution.

2.2 Factor-Based Identification and Sampling Framework

We model untreated outcomes as

$$Y_{it}(0) = X_{it}'\beta_0 + \lambda_i' f_t + h_0(X_{it}, \lambda_i, f_t) + \varepsilon_{it}, \tag{3}$$

where X_{it} contains observed characteristics, f_t captures common changes, and λ_i describes how unit i responds to them. The function h_0 lets units respond nonlinearly to the same

factors and covariates without introducing additional factors; setting $h_0 \equiv 0$ gives GSC's linear interactive fixed effects (IFE) model. The covariate distribution may change, but the response function remains common across groups and time. Observed outcomes are $Y_{it} = Y_{it}(0) + \delta_{it} D_{it}$, where the treatment effect δ_{it} may differ across units and periods.

To compare treated and control units, we use adjustment information that combines factor loadings with summaries of pre-treatment covariates, outcomes, and linear-model residuals. Because prediction also depends on conditions in the period of interest, the prediction inputs include common time factors and contemporaneous covariates. The following assumptions state when this information identifies the missing untreated outcomes; formal definitions and technical conditions are in SI, Appendix A.

Assumption 1 (SUTVA)**.** *Each unit's outcome depends on its own treatment status, not on the treatment status of other units.*

Assumption 2 (Anchored nonlinear factor structure)**.** *Untreated outcomes follow equation* (3)*. The linear factor component is strong, and the nonlinear response uses the same covariates, factor loadings, and common factors. Conditional on these inputs, the disturbance has mean zero.*

Assumption 3 (Factor-based unconfoundedness)**.** *Among units with the same adjustment information, treated and control units share the same joint distribution of untreated outcome paths and observed covariates used for prediction. These covariates are predetermined or unaffected by treatment. Together with the common time factors, they provide no further information about treatment assignment.*

This allows controls with the same adjustment information to represent the treated group's untreated outcome distribution.

Assumption 4 (Overlap condition)**.** *At each value of the adjustment information, both treated and control units occur with probabilities bounded away from zero.*

Assumption 5 (Regularity conditions)**.** *Whole unit histories are independent across units conditional on the common design environment, while outcomes within a unit may be dependent across post-treatment periods. Outcomes satisfy the moment and pre-treatment concentration conditions in Appendix A.*

Assumption 6 (Recoverability of the linear factor component)**.** *The linear factor component can be distinguished from the remaining variation. As the control group and pre-treatment window grow, the fitted factors and loadings recover the information needed for adjustment. If the number of factors is selected, the probability of selecting the correct number approaches one, under the stated rank, separation, and projection conditions.*

3 Method: DR-FGAN for ATT and ETC-GAN for Marginal QTE

3.1 Overall Design

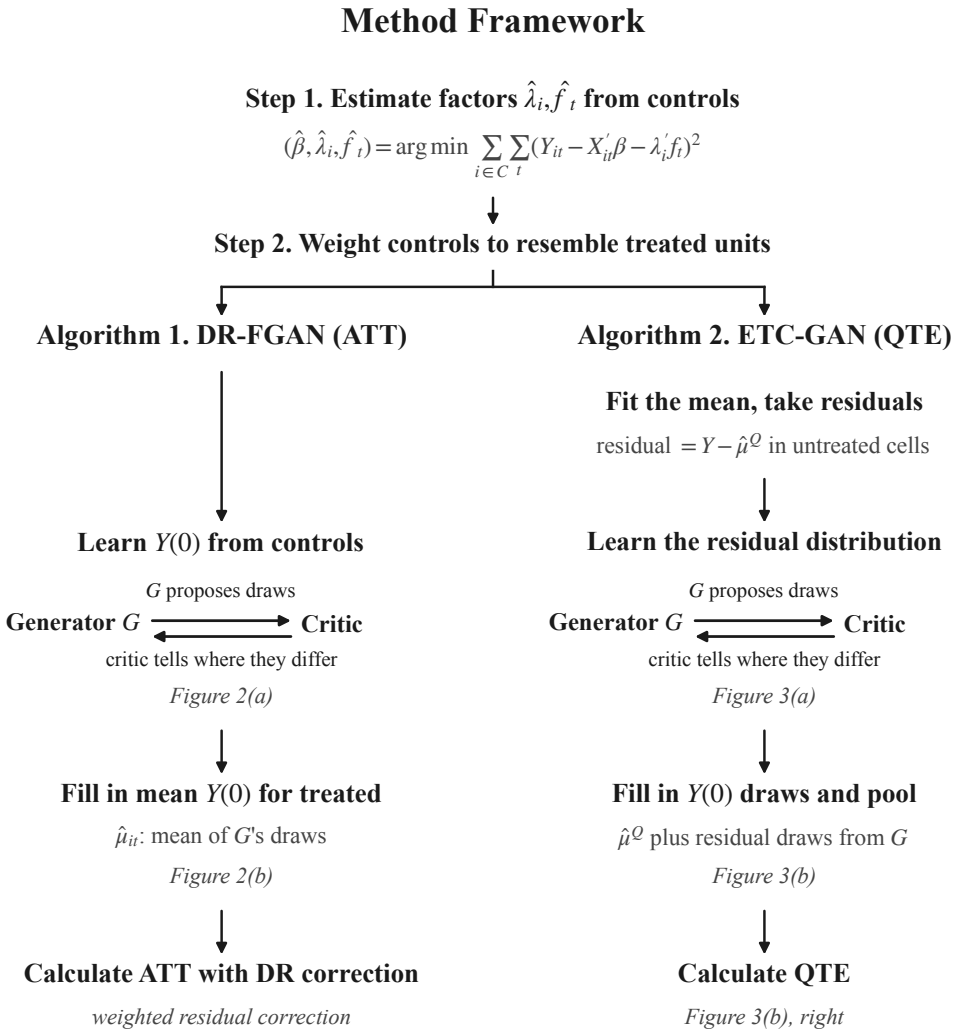


Figure 1: Method framework. The estimators share factor adjustment and control weights, then learn untreated means for ATT or untreated distributions for QTE.

G: Outcome generator in DR-FGAN; residual generator in ETC-GAN
Y_{it}: Observed outcome
$Y(0)$: Outcome without treatment
C: Training control units
X_{it}: Observed covariates
β: Covariate coefficients
i, t: Unit and period indices
$\hat{\lambda}_i$: Estimated unit loadings
$\hat{f}_t$: Estimated common factors
$\hat{\mu}_{it}$: DR-FGAN untreated mean
$\hat{\mu}^Q$: Separate QTE regression mean
ATT: Average treatment effect on treated units
QTE: Quantile treatment effect
DR-FGAN: Doubly robust factor-augmented GAN
ETC-GAN: Empirical-transport Cramér GAN
GAN: Generative adversarial network
DR: Doubly robust

Notes: Hats denote estimates; primes denote transposes.

Figure 1 begins with two shared steps: fitting the linear factor model and reweighting controls to resemble treated units. The factor fit recovers common time factors and unit-specific loadings, which become prediction inputs; the weights use the adjustment information to make controls comparable to treated units.

The two branches then learn different features of the untreated outcomes. DR-FGAN averages its generator's draws to predict the untreated mean, then corrects the treated group's observed–predicted gap using weighted control residuals. ETC-GAN fits a separate mean and learns the residual distribution. It adds the mean back to the generated residuals and pools the resulting outcomes before comparing quantiles. In each learning loop, a critic identifies differences between generated and observed untreated outcomes and guides the generator's updates. Figures 2 and 3 show these steps over time. SI, Appendix B, gives the fitting algorithms, weight formulas, training objectives, and network specifications for this section.

3.2 Stage 1: Estimating the Linear Factor Model

Following Xu (2017), we estimate the linear factor model from control outcomes. We split whole units into folds, leaving one fold out at a time. Controls in the remaining folds estimate the common factors over all periods; each held-out unit's loadings come from its own pre-treatment outcomes. When the number of factors is not fixed in advance, we choose it by predicting treated units' pre-treatment outcomes with one period left out at a time.

3.3 Stage 2: Propensity Score Estimation and IPW Reweighting

We estimate each unit's probability of treatment, $\hat{e}_i$, from its adjustment information, using a classifier trained on other units. The simulations and electricity application use gradient-boosted trees, while the political-connections application uses standardized regularized logistic regression, a lower-complexity model for its 34 connected firms. Both branches share the resulting scores and folds.

Control weights start proportional to the estimated treatment odds, $\hat{e}_i/(1-\hat{e}_i)$, so controls whose characteristics resemble those of treated units receive more weight. To keep a few controls from dominating the fit, we clip extreme propensity scores and cap large weights, then normalize the final control weights, $\tilde{w}_i$, to sum to one.

3.4 Stage 3A: Learning the Untreated Mean for ATT

DR-FGAN learns the untreated mean with a Wasserstein GAN with gradient penalty (WGAN-GP; Gulrajani et al. 2017). Its generator G builds on the linear factor prediction and adds a nonlinear correction that depends on the observed covariates, estimated loadings, and common factors. The loadings and factors stay fixed during training, while random inputs let the generator produce many draws for the same unit and period. A critic, trained in alternation with the generator, scores pairs of predictors and outcomes and learns to distinguish observed control outcomes from generated ones, so the generator improves by making the two harder to distinguish. Both sets use the same control weights.

The critic is not the only guide. The generator is also rewarded for fitting observed control outcomes and matching treated units' pre-treatment means, and is penalized for moving far from the linear factor prediction. We keep the training round with the smallest pre-treatment prediction error for held-out treated units; the fitted untreated mean $\hat{\mu}_{it}$ averages that generator's draws for each unit and period. Figure 2 traces these steps: panel (a) shows the generator moving from the linear factor prediction toward control outcomes, and panel (b) applies it to treated units.

3.5 Stage 3B: Learning the Residual Distribution for QTE

For quantile effects, we first fit a separate regression for the untreated mean, $\hat{\mu}_{it}^{Q}$, and then learn how outcomes vary around it. The regression adds a nonlinear correction to the linear factor prediction and minimizes weighted squared error. For untreated observations, predictions from other unit folds give residuals $\hat{r}_{it} = Y_{it} - \hat{\mu}_{it}^{Q}$.

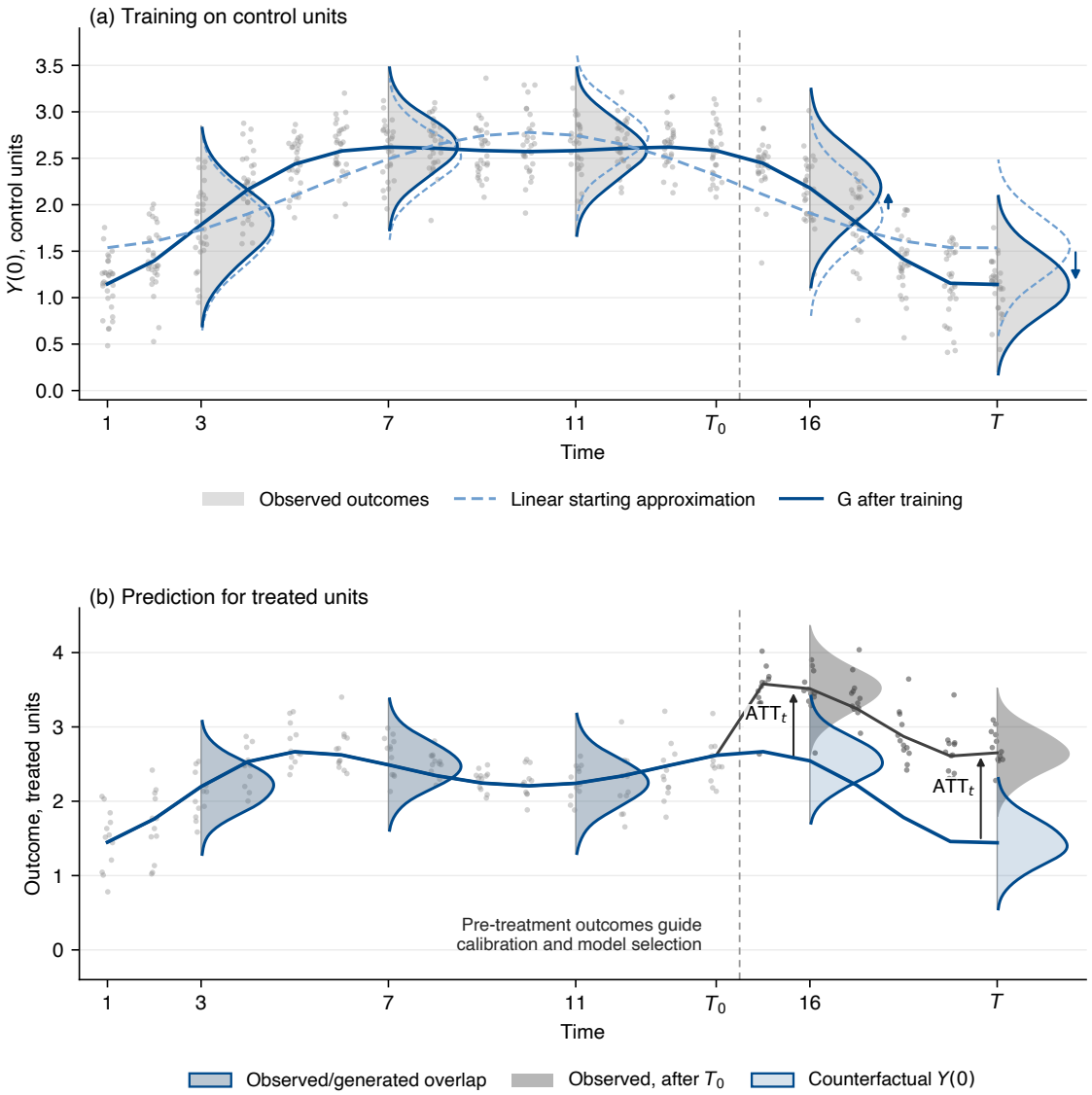


Figure 2: DR-FGAN: learning untreated outcomes and comparing group means.

G: Outcome generator
$Y(0)$: Outcome without treatment
t: Period index
T_0: Last pre-treatment period
T: Final observed period
ATT_t: Average treatment effect on treated units in period t
DR-FGAN: Doubly robust factor-augmented generative adversarial network

Notes: Schematic. Dots represent unit-period observations, sideways shapes show densities across units, and lines connect group means. In (a), G adjusts the linear prediction toward observed controls. In (b), overlap shows agreement before treatment; after T_0, gray shows observed treated outcomes and blue shows predicted untreated outcomes. The arrows mark the mean gap before the weighted residual correction in equation (4).

Holding this mean fixed, ETC-GAN starts from the weighted empirical residual quantiles. Its separately fitted residual generator G adjusts their positions and spacing for each set of prediction inputs, keeping the quantiles in order. Drawing a percentile uniformly between zero and one selects a residual, so many draws reproduce the distribution rather than just its mean. A tail-weighted quantile loss and an energy-score criterion compare these draws directly with observed residuals. A Cramér critic learns additional features on which the generated and observed distributions differ. Figure 3 illustrates how the shared starting distribution adapts to different factor inputs: the generator changes the location, spread, and tails of the residual distribution. Adding a generated residual $\tilde{r}_{it}$ to the regression mean gives an untreated outcome draw,

$$\tilde{Y}_{it}(0) = \hat{\mu}_{it}^{Q} + \tilde{r}_{it},$$

as shown in panel (b). Pooling these draws across the treated units and target periods gives the counterfactual distribution whose quantiles are compared with the observed distribution in the right-hand panel.

3.6 Stage 4: Counterfactual Aggregation for ATT and QTE

3.6.1 Doubly robust ATT estimation

The doubly robust ATT estimator is:

$$\widehat{ATT}_t^{\mathrm{DR}} = \frac{1}{N_{\mathrm{tr}}} \sum_{i \in \mathcal{T}} \big(Y_{it} - \hat{\mu}_{it}\big) - \sum_{i \in \mathcal{C}} \tilde{w}_i \big(Y_{it} - \hat{\mu}_{it}\big). \tag{4}$$

The first term contains the treatment effect plus the prediction error for treated units. The second estimates the corresponding error from controls, reweighted to resemble the treated group, and subtracts it.

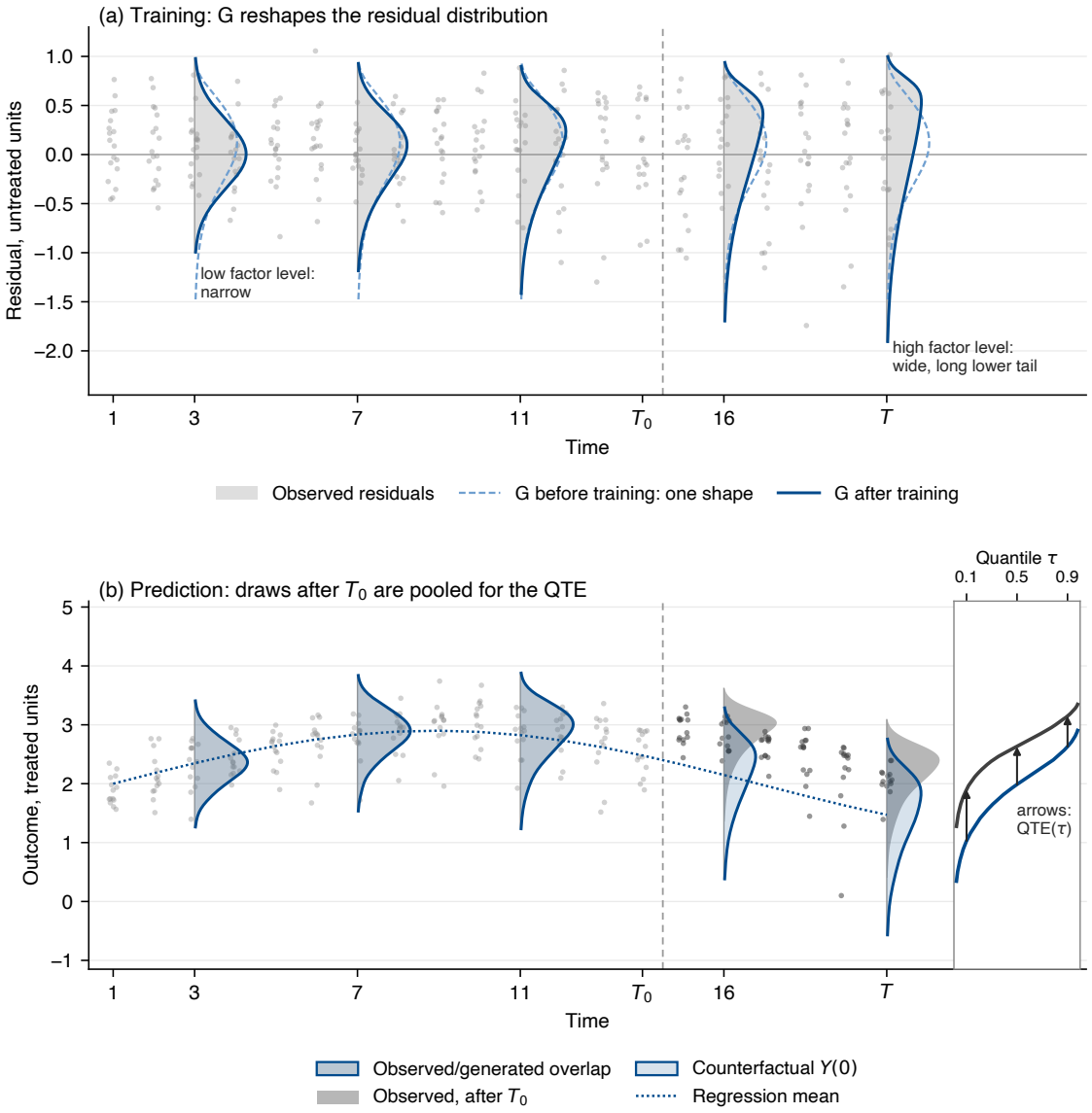


Figure 3: ETC-GAN: learning residual distributions and comparing quantiles.

G: Residual generator
$Y(0)$: Outcome without treatment
T_0: Last pre-treatment period
T: Final observed period
τ: Quantile level, between zero and one
$QTE(\tau)$: Quantile treatment effect: treated minus counterfactual quantile at level τ
ETC-GAN: Empirical-transport Cramér generative adversarial network

Notes: Schematic. Dots are unit-period residuals in (a) and outcomes in (b); sideways shapes show densities across units. In (a), G adapts residuals around the fitted untreated mean to the factor inputs. In (b), adding the regression mean (dotted) gives the blue untreated distributions; gray shows observed outcomes. Overlap denotes agreement before treatment. The inset pools outcomes over treated units and post-treatment periods, then compares their quantiles at $\tau = 0.1$, 0.5, and 0.9.

3.6.2 Distributional estimation and QTE

To estimate the marginal QTE in equation (2), we first pool the generated untreated outcomes across treated units in period t. We then compare each percentile of this counterfactual distribution with the same percentile of the observed treated outcomes, as in Figure 3(b). To pool across post-treatment periods, we average the period-specific distributions, with equal weights in our applications, and then take quantiles. Averaging period-specific QTEs would answer a different question. The estimate is called plug-in because it replaces the unknown counterfactual distribution with the fitted distribution before comparing quantiles. For a single target period t, we write $\widehat{QTE}_t^{\text{plug}}(\tau)$ for this estimate at quantile level τ.

4 Theoretical Properties

This section states the main results. The formal recovery, learning, and inference conditions and all proofs are in SI, Appendix A.

4.1 Deep Learning and Conditional Mean Consistency

Assumption 7 (Consistency of propensity score estimation)**.** *With overlap and consistent recovery of the adjustment information, estimated treatment probabilities converge uniformly to the true probabilities conditional on that information.*

Assumption 8 (Sufficiency of the critic function class)**.** *The critic class distinguishes the weighted observed and generated distributions: a vanishing population Wasserstein discrepancy implies equality of those distributions.*

Theorem 1 (Conditional mean consistency)**.** *Under the recovery and mean-learning conditions in SI, Appendix A, the fitted untreated mean $\hat{\mu}_{it}$ consistently estimates the untreated conditional mean: its population mean-squared error relative to that mean converges to zero in probability.*

The prediction approaches the average untreated outcome for units with the same observed characteristics, factor loadings, and common factors.

4.2 Double Robustness

The ATT correction uses control residuals to estimate prediction error in the treated population. We first define consistency for each fitted function.

Condition 1 (Consistent outcome prediction)**.** *The fitted untreated mean $\hat{\mu}_{it}$ converges in mean square to the untreated mean conditional on the prediction inputs, under the stated evaluation conditions.*

Condition 2 (Consistent propensity score)**.** *Estimated treatment probabilities converge uniformly to the true probabilities conditional on the adjustment information.*

Theorem 2 (Double robustness)**.** *Under the adjustment and estimation conditions in SI, Appendix A:*

(a) If the fitted outcome prediction is consistent, then $\widehat{ATT}_t^{\mathrm{DR}} \xrightarrow{p} ATT_t$, allowing a misspecified propensity limit.

(b) If the propensity score is consistent, the same conclusion holds with a possibly misspecified outcome limit of finite second moment.

(c) If both are consistent and the additional inference conditions hold, the estimator has asymptotically normal estimation error and, to first order, the same sampling behavior as the corresponding estimator using the true adjustment information, untreated mean, and treatment probabilities.

The correction thus offers two routes to consistent ATT estimation: accurate outcome predictions or accurate treatment probabilities. The bias decomposition in the SI shows why weighting matters: after weighting, control residuals measure the prediction error that treated units would have had without treatment. The same identity explains the shift design in Section 5.3.

4.3 Behavior under a Linear DGP

We write LFM for the cross-fitted linear factor prediction that DR-FGAN starts from, used without the neural correction or the residual correction.

Proposition 1 (Population limit under a linear DGP)**.** *If $h_0 \equiv 0$ and the SI's linear-limit conditions hold, DR-FGAN and LFM have the same conditional-mean limit.*

In other words, when the true response is linear and these conditions hold, DR-FGAN and the linear factor estimator converge to the same prediction.

4.4 ATT Variance Estimation

Proposition 2 (Conditional variance estimation)**.** *Under the variance-estimation conditions in SI, Appendix A, the unit-cluster score method consistently estimates the ATT estimator's first-order variance. The delete-one-unit method does so under the additional deletion-stability conditions stated there.*

Both methods treat a unit's full history as one observation, allowing outcomes within that history to be related over time.

In the political-connections application, we resample whole firms within treatment-by-fold strata and refit the entire estimator on each sample. The bootstrap theory covers the sampling variation of this full procedure, including factor recovery and rank selection, under its stated conditions.

4.5 Plug-in Marginal-QTE Consistency

Proposition 3 (Consistency of the plug-in QTE)**.** *Let $\mathcal{A}$ be a fixed closed set of quantile levels inside $(0, 1)$. Under the identification, recovery, and distribution-learning conditions in SI, Appendix A,*

$$\sup_{\tau \in \mathcal{A}} \left| \widehat{QTE}_t^{\text{plug}}(\tau) - QTE_t(\tau) \right| = o_p(1).$$

The conclusion also applies when distributions from a fixed set of periods are combined before taking quantiles, under the corresponding mixture conditions.

The estimated differences between corresponding percentiles approach the true differences across this range. The proof follows the estimator: learn residual distributions, add the conditional mean, pool over treated units, and compare quantiles.

5 Monte Carlo Simulations

We use one simulation design for ATT, with twenty replications, and another for quantile effects, with thirty replications. We then change the covariate distribution in the ATT design and estimate both effects on the shifted panels. Full specifications and comparisons are in SI, Appendix C.

5.1 ATT benchmark: Nonlinear responses without covariate shift

Design. Baseline vulnerability Z_i interacts nonlinearly with an external covariate W_{it}, while latent factors generate comovement:

$$\begin{aligned} Y_{it}(0) &= 0.7Z_i + 0.5W_{it} + 1.5\lambda_{i1}f_{1t} + 1.2\lambda_{i2}f_{2t} + h(Z_i, W_{it}) - a(Z_i) + \varepsilon_{it}, \\ h(z,w) &= 3\tanh(1.2z)\tanh\{1.2(z+w)\}, \qquad a(z) = \mathbb{E}_{W\sim U[-2,2]}\{h(z,W)\}. \end{aligned} \tag{5}$$

The population function $a(z)$ centers the nonlinear response: under the original distribution of W, this response averages to zero at each value of Z. Nonlinear prediction errors can therefore cancel in the ATT even when a linear model misses individual outcomes. Section 5.3 changes the distribution of W, disrupting this cancellation. The design lets the linear factor component be estimated separately from the nonlinear term.

We draw $Z_i \sim U[-1.5, 1.5]$, independent $\lambda_{ik} \sim U[-\sqrt{3}, \sqrt{3}]$, and independent Gaussian noise with SD 0.3. Treatment is assigned once with probability $\text{expit}(c+Z_i)$, where c fixes the population treated share at 0.35. Thus selection depends on observed vulnerability, while

Table 1: ATT accuracy when the covariate distribution is stable

Method	Bias	MAE	SD(error)	RMSE
DR-FGAN	**-0.00378**	**0.04260**	**0.05367**	**0.05245**
ASC	0.05363	0.11118	0.12456	0.13273
DID/TWFE	0.03517	0.13259	0.17506	0.17422
SDID	0.00572	0.05442	0.07258	0.07097

ATT: Average treatment effect on the treated
GAN: Generative adversarial network
DR-FGAN: Doubly robust factor-augmented GAN
ASC: Augmented synthetic control
DID/TWFE: Difference-in-differences with two-way fixed effects
SDID: Synthetic difference-in-differences
MAE: Mean absolute error
RMSE: Root mean squared error
SD(error): Sample standard deviation of estimation errors
N: Number of units
T: Number of periods
T_0: Number of pre-treatment periods
SI: Supplementary Information

Notes: Entries summarize twenty replications with $N = 300$, $T = 48$, $T_0 = 36$, and true ATT 2. The external covariate has the same distribution before and after treatment. Bias records average over- or underestimation; MAE and RMSE summarize error size, while SD(error) measures variation across replications. Smaller MAE and RMSE and bias closer to zero indicate greater accuracy. DR-FGAN combines covariates, factor inputs, and residual correction; ASC, DID/TWFE, and SDID use outcome histories without added observed covariates. SDID reaches its default 10,000-iteration limit in 20 of 20 replications. The SI gives complete comparisons, training diagnostics, and paired Monte Carlo uncertainty.

latent loadings contribute to outcome heterogeneity. There are $N = 300$ units, $T_0 = 36$ pre-treatment periods, and twelve post-treatment periods. Treatment adds two to every treated post-treatment outcome. Population and realized pooled ATT equal two, as do all period-specific and pooled marginal QTEs.

Here $W_{it} \sim U[-2, 2]$ in every period and both groups: values vary, but the distribution is unchanged.

Estimation and comparison. Table 1 compares DR-FGAN with ASC, DID/TWFE, and SDID. DR-FGAN uses observed covariates and fixed factor estimates, with three unit folds and pre-treatment rank selection; all twenty panels select rank two. ASC uses ridge augmentation of pre-treatment outcome histories; DID/TWFE includes unit and time effects; SDID combines unit and time weighting. All methods use the same twenty panels.

Results. DR-FGAN's RMSE is 0.052, approximately 60 percent below ASC's (0.133), 70 percent below DID/TWFE's (0.174), and 26 percent below SDID's (0.071). Its absolute bias

is also smaller. DR-FGAN uses Z_i and W_{it} directly, so it can learn that the response to W depends on Z. The three benchmarks work from outcome histories without these observed covariates. The cross-fitted linear factor prediction (LFM), included in the SI comparison, has a similar RMSE here (0.053), as expected when the centered nonlinear errors cancel in the ATT.

5.2 QTE benchmark: Nonlinear factors and heterogeneous distributional effects

Design. The distribution benchmark uses $N = 180$, $T_0 = 30$, and ten post-treatment periods. Covariates and factors have independent Gaussian draws without a pre/post distribution change. Untreated means depend nonlinearly on the factor index and covariate interactions; variances and negative-shock probabilities also vary with latent state. Selection depends on loadings and pre-treatment covariate averages.

Writing $Y_{it}(0) = \mu_{it} + \sigma_{it}\varepsilon_{it}$, treatment cushions negative shocks in treated post-treatment outcomes:

$$Y_{it}(1) = \mu_{it} + \sigma_{it}\{\varepsilon_{it} + 0.85 \max(0, -\varepsilon_{it})\}.$$

Treatment reduces the magnitude of negative shocks by 85 percent while leaving positive shocks unchanged. The design therefore emphasizes protection against adverse outcomes, making the lower part of the distribution particularly informative. The target compares quantiles of the treated cohort's pooled post-treatment potential outcomes, with effects that vary by quantile.

Estimation and comparison. Table 2 uses thirty replications, a fixed rank of four, and two unit folds. ETC-GAN learns a conditional residual distribution and combines it with the regression mean to reconstruct untreated outcomes. Pooling these outcomes gives the counterfactual quantiles used in the QTE plug-in.

Group-level CiC (Athey and Imbens 2006) locates each treated pre-period outcome in

the control pre-period distribution, then maps it to the control post-period outcome at that same percentile. Group-level CDF-DiD (Roth and Sant'Anna 2023) adds the control group's change in the outcome CDF to the treated group's pre-period CDF at every threshold. Both use the same thirty panels.

Table 2: Accuracy of quantile treatment effect estimates

	Pooled QTE MAE				Distribution fit	
Method	Overall (.05–.95)	Lower (.05–.30)	Central (.35–.65)	Upper (.70–.95)	CRPS	Calibration (PIT)
ETC-GAN	**0.1831**	**0.2285**	**0.1266**	**0.2036**	**0.9389**	**0.0388**
Changes-in-changes (group level)	0.2025	0.2690	0.1614	0.1840	–	–
CDF-DiD (group level)	0.2039	0.2727	0.1605	0.1857	–	–

QTE: Quantile treatment effect
GAN: Generative adversarial network
ETC-GAN: Empirical-transport Cramér GAN
CDF: Cumulative distribution function
CDF-DiD: CDF difference-in-differences
MAE: Mean absolute error
CRPS: Continuous ranked probability score
PIT: Probability integral transform
N: Number of units
T: Number of periods
T_0: Number of pre-treatment periods
SI: Supplementary Information

Notes: QTEs compare matching quantiles of treated outcomes and reconstructed untreated outcomes, each pooled over treated units and post-treatment periods. The group-level benchmarks transform pooled outcome distributions rather than fitting a distribution for each unit and period. The first four columns give average absolute errors: Overall averages $.05, .10, \ldots, .95$; Lower, Central, and Upper average .05–.30, .35–.65, and .70–.95. CRPS measures how well predicted distributions fit simulated untreated outcomes. PIT error measures percentile calibration: for example, about 25% of these outcomes should fall below their predicted 25th percentiles. Lower scores indicate better fit or calibration. The conditional-distribution scores do not apply to the group-level benchmarks and are shown as dashes. Bold identifies the proposed estimator. Entries average thirty panels with $N = 180$, $T = 40$, and $T_0 = 30$ under the nonlinear, asymmetric-error design in Section 5.2. The distributions of covariates and factors do not change at the treatment date. The learner uses a pre-specified rank of four and two unit folds. The SI gives design and training specifications and the complete five-method comparison.

Results. ETC-GAN's overall QTE error is 0.183, about 10 percent below group-level CiC (0.203) and CDF-DiD (0.204). The largest reduction is in the lower region: 0.229, compared with 0.269 and 0.273. This region is important because treatment in this design cushions negative shocks. ETC-GAN learns how the residual distribution changes with each unit's covariates and factors, whereas the group-level methods transform pooled outcome distributions.

5.3 Extension: Changing the external covariate distribution

We change only the ATT design's post-treatment distribution of W to $0.35U[-2,0] + 0.65U[0,2]$ in both groups, pairing the same twenty seeds with the no-shift benchmark. Treatment does not cause W; support and the response function remain unchanged. In the nonlinear term $h(Z,W)$, the response rises with W for positive-Z units and falls for negative-Z units. Treated units have higher Z on average. Because positive values of W become more common after treatment, treated units' untreated outcomes rise relative to the controls'. A linear factor prediction that misses this nonlinear change leaves part of the increase in the estimated treatment effect.

GSC's bias is 0.223 and its RMSE is 0.233 (Table 3). The positive biases of LFM and GSC mean that they attribute part of the covariate-driven increase in untreated outcomes to treatment. DR-FGAN's RMSE is 0.052, about 78 percent lower, with bias of -0.003. Its RMSE is also about 75 percent below LFM's (0.208), and its absolute error is smaller than both methods' in every paired panel. Learning the nonlinear response matters here because the change in W affects the two groups differently.

QTE estimation uses these same twenty shift panels, with three unit folds and pre-treatment rank selection over one through four. All panels select rank two. ETC-GAN's overall QTE MAE is 0.064, compared with 0.291 for CiC and 0.284 for CDF-DiD. It has smaller error in all twenty panels. Conditioning on covariates and factors allows its predictions to change with the current covariates of the treated units.

6 Empirical Applications

We apply both estimators to political connections during the 2008 financial crisis and, as a supplement, to a dynamic electricity tariff in London.

Table 3: ATT accuracy when the covariate distribution shifts

Method	Bias	MAE	SD(error)	RMSE
DR-FGAN	**-0.00334**	**0.04230**	**0.05285**	**0.05162**
LFM (linear factor prediction)	0.20090	0.20090	0.05443	0.20779
GSC	0.22342	0.22342	0.06860	0.23321
ASC	0.06660	0.11222	0.12854	0.14189
DID/TWFE	0.25787	0.25885	0.17460	0.30897
SDID	0.25593	0.25593	0.08126	0.26791

ATT: Average treatment effect on the treated
GAN: Generative adversarial network
DR-FGAN: Doubly robust factor-augmented GAN
LFM: Linear factor model
GSC: Generalized synthetic control
ASC: Augmented synthetic control
DID/TWFE: Difference-in-differences with two-way fixed effects
SDID: Synthetic difference-in-differences
MAE: Mean absolute error
RMSE: Root mean squared error
SD(error): Sample standard deviation of estimation errors
SI: Supplementary Information

Notes: The twenty replications share seeds, sample sizes, and estimation specifications with Table 1; only the post-treatment covariate distribution changes. True ATT remains 2. Bias records average over- or underestimation; MAE and RMSE summarize error size, while SD(error) measures variation across replications. Smaller MAE and RMSE and bias closer to zero indicate greater accuracy. LFM uses the cross-fitted linear factor prediction underlying DR-FGAN, without its neural adjustment or weighted control-residual correction. GSC is fitted separately with the official `gsynth` implementation and its own rank selection. SDID reaches its default 10,000-iteration limit in 19 of 20 replications. The SI gives complete comparisons, training diagnostics, ablations, and paired Monte Carlo uncertainty.

6.1 Political connections during the financial crisis

On November 21, 2008, news leaked that Timothy Geithner, then president of the Federal Reserve Bank of New York, would be nominated U.S. Treasury Secretary. Acemoglu et al. (2016) find that financial firms connected to Geithner gained in value after the news. We revisit this event using their replication data, defining connections through personal ties or meetings recorded in his schedule. Financial firms move with the market but differ in how strongly they move with it; the factor model captures this difference. The distributional analysis asks whether the return contrast is larger in the lower part of the distribution than near the middle.

6.1.1 Data and research design

We calculate daily percentage returns from total-return indices at consecutive trading-day closes. The window includes $T_0 = 100$ pre-event trading days, July 2–November 20, 2008, and ten post-event trading days, November 21–December 5. We exclude the market index and firms with incomplete observations, and require nonzero returns on more than half of the 100 pre-event dates. This leaves 622 firms: 34 connected firms and 588 controls. Adjustment uses pre-event returns and estimated factors (data processing and diagnostics are in SI, Appendix D).

The ATT is the average daily effect for the eligible connected-firm population at the event dates; we also report the sum of daily effects. QTE compares the pooled firm–day return distributions with and without political connections over the same window.

We assign whole firms to two outer folds and fit the linear factor model using training controls. Pre-event leave-one-period-out validation selects rank six from the grid $1, \ldots, 10$; a held-out firm's loading uses its own pre-event returns. Both branches share the linear factor prediction, fold assignments, propensity predictions, and stabilized control weights, but learn their outcome predictions separately. The adjustment information combines estimated loadings with summaries of pre-event returns and linear-model residuals.

For confidence intervals, we draw 199 bootstrap samples of whole firms, resampling separately within each combination of connection status and fold. On each sample, we reselect rank and refit both estimators. The event dates and training settings remain fixed.

6.1.2 Average return effects

Panel A of Table 4 reports DR-FGAN estimates after weighted control-residual correction. The average daily estimate is 0.912 percentage points and the ten-day sum is 9.120 percentage points. The 95% bootstrap normal-approximation intervals are $[-0.106, 1.930]$ for the daily average and $[-1.060, 19.299]$ for its ten-day sum. Both point estimates are positive, but their intervals include zero.

Table 4: Average and distributional return effects of Geithner connections

Panel A. DR-FGAN average effects (percentage points)				
Estimand	Estimate			95% interval
Mean daily ATT	0.912			[-0.106, 1.930]
10-day sum	9.120			[-1.060, 19.299]
Panel B. Plug-in ETC-GAN marginal quantile effects				
Percentile	Observed (%)	Counterfactual (%)	QTE (pp)	95% interval
25th	-0.522	-3.865	3.343	[1.064, 3.640]
50th (median)	3.309	1.432	1.876	[0.047, 2.757]
75th	7.935	7.105	0.830	[-0.181, 1.835]

ATT: Average treatment effect on the treated
QTE: Quantile treatment effect
DR-FGAN: Doubly robust factor-augmented generative adversarial network
ETC-GAN: Empirical-transport Cramér generative adversarial network
pp: Percentage points
SI: Supplementary Information

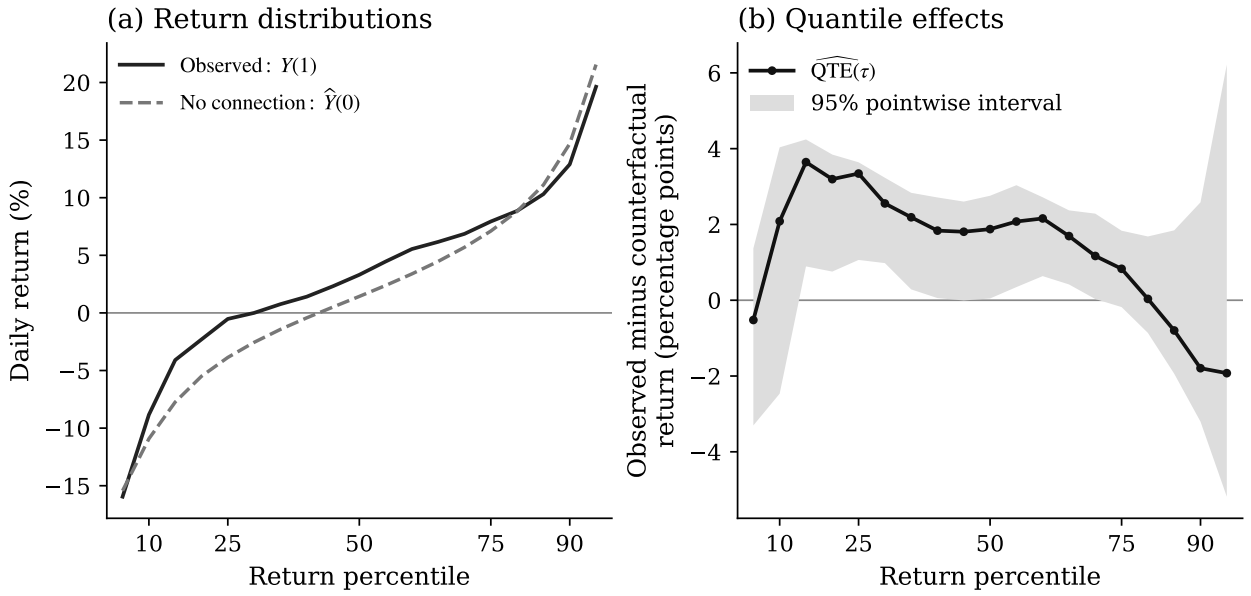
The sample contains 622 firms: 34 connected firms and 588 controls, over 100 pre-event and ten post-event trading dates. Panel A reports effects on daily returns in percentage points; the ten-day sum adds daily effects rather than compounding returns. Panel B compares the same percentiles of observed and counterfactual daily returns pooled over 340 connected firm-days. Whole firm histories are resampled within groups defined by treatment status and estimation fold. Both estimators use the same 199 resamples; each time, rank is reselected from 1–10 and the factors, weights, and networks are re-estimated. The 95% intervals use a normal approximation for ATT and pointwise percentiles for QTE. The event calendar and treatment-by-fold counts are fixed. Data-processing rules, bootstrap assumptions, and resampling diagnostics appear in the SI.

6.1.3 Distributional effects

For each of the 34×10 post-event firm–days, we combine the predicted mean with the fitted residual distribution. Averaging these distributions with equal weights gives the estimated no-connection distribution, whose quantiles we compare with observed returns.

Figure 4 compares observed and counterfactual return quantiles. Table 4 reports the three quartile contrasts and their 95% pointwise bootstrap percentile intervals.

The largest of the three contrasts is at the lower quartile. The 25th percentile of connected firms' daily returns is -0.522 percent, compared with -3.865 percent in the estimated no-connection distribution: a gap of 3.343 percentage points, with a 95% interval of $[1.064, 3.640]$. The gaps at the median and upper quartile are smaller, at 1.876 and 0.830 percentage points. The average alone therefore misses a larger contrast among lower returns.



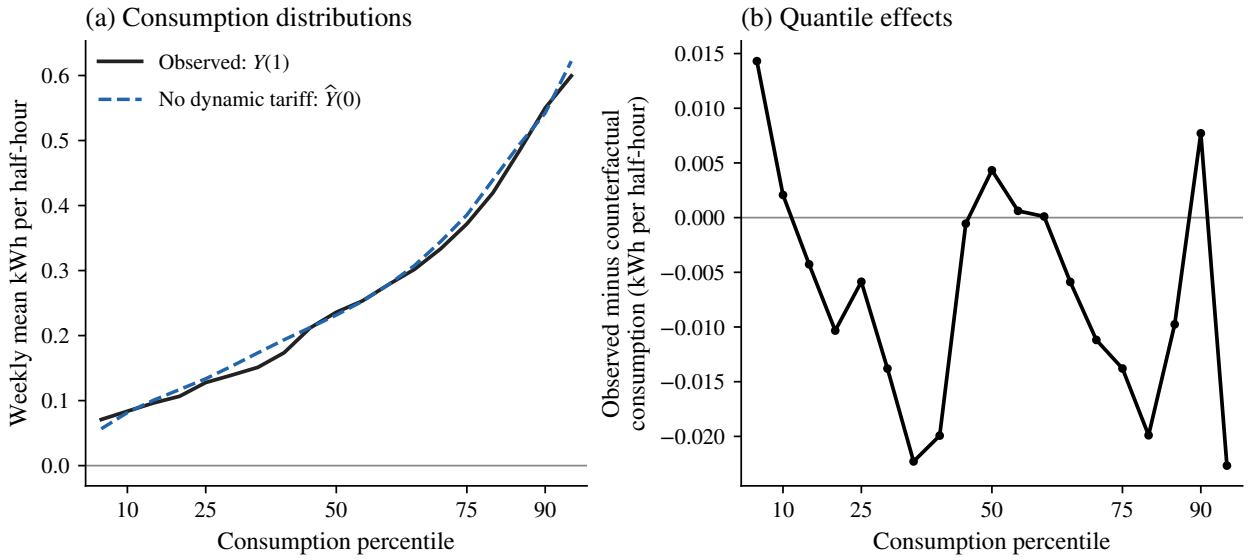
ETC-GAN: Empirical-transport Cramér generative adversarial network
QTE: Quantile treatment effect
$Y(1)$: Observed connected-firm returns
$\widehat{Y}(0)$: Estimated returns without connections
$\widehat{Q}_1, \widehat{Q}_0$: Observed and counterfactual quantile functions
τ: Quantile level

Figure 4: Observed returns and estimated returns without political connections. Panel (a) compares observed (solid) and counterfactual (dashed) quantiles, pooling 34 connected firms over ten post-event days. Panel (b) shows their difference in percentage points. Shading gives 95% pointwise percentile intervals from 199 whole-firm bootstrap refits; the horizontal line marks zero.

6.2 Supplementary application: London electricity tariffs

The Low Carbon London smart-meter panel records household consumption and tariff participation (UK Power Networks 2022). We link these records to weather observations from NOAA (NOAA National Centers for Environmental Information 1999). Dynamic tariffs began on January 4, 2013, with day-ahead notifications. Recruitment and incentive offers preceded operation (UK Power Networks 2012; 2013a). Both groups received usage and cost feedback; tariff participants additionally received a rate breakdown (UK Power Networks 2013b). The estimand is the effect of this operating-period tariff bundle relative to the standard tariff.

We use 52 pre-operation weeks in 2012 and 13 post-operation weeks from January 7 to April 7, 2013, omitting the mixed transition week. From households with near-complete readings in both periods, we draw a random sample of 36 tariff participants and 108 controls,



$Y(1)$: Observed participant consumption
$\widehat{Y}(0)$: Estimated consumption without dynamic tariffs
$\widehat{Q}_1, \widehat{Q}_0$: Observed and counterfactual quantiles
τ: Quantile level

ETC-GAN: Empirical-transport Cramér generative adversarial network
QTE: Quantile treatment effect
kWh: Kilowatt-hour

Figure 5: Electricity consumption with and without the dynamic tariff. Panel (a) compares observed and ETC-GAN counterfactual quantiles, pooling 36 households over 13 operating weeks with equal weights. Panel (b) shows their difference; the gray line marks zero. Both panels show point estimates at rank three, selected from $\{1, 2, 3\}$. Units are weekly mean kWh per valid half-hour (observed median 0.236).

out of 52 and 284 eligible households, respectively. For each household and week, we average consumption across its valid half-hour readings, giving weekly mean kWh per valid half-hour. The baseline characteristic Z_i is mean pre-period consumption, and W_t is Heathrow's weekly temperature, common to all households. The winter post-period is colder: mean temperature falls from 11.04 to 3.74 degrees Celsius. Tariff participation does not affect the weather, so consumption can change even without a tariff change. GSC's time effects absorb the common temperature series. The networks can also learn nonlinear temperature responses that depend on baseline consumption and estimated household characteristics.

Both estimators use the same three whole-household folds and factor estimates. They also share propensity scores, but fit their outcome models separately. Validation over $\{1, 2, 3\}$ selects rank three. DR-FGAN estimates ATT at -0.00106 kWh per valid half-hour. ETC-GAN pools counterfactual draws with equal household and week weights. Figure 5 compares

observed and no-dynamic-tariff distributions. Lower-quartile, median, and upper-quartile QTEs are -0.00586, 0.00433, and -0.01379 (SI, Appendix E, for rank and initialization comparisons). Relative to the pooled observed median of 0.236 kWh, these point estimates are small, each below 6 percent of that median in absolute value.

7 Discussion and Conclusion

The results suggest a practical reason to compare linear and nonlinear predictions. In our stable-distribution design, the centered nonlinear response largely cancels in ATT, so DR-FGAN and linear factor estimators give similar answers. When the covariate distribution changes, this cancellation breaks down: the linear factor prediction attributes part of the change in untreated outcomes to treatment, while DR-FGAN removes most of this error. Comparing the estimates can therefore help assess how much the answer depends on a linear response.

Average and distributional effects answer different substantive questions. DR-FGAN corrects the aggregate mean using weighted control residuals, while ETC-GAN learns the residual distribution needed for quantiles. In the political-connections application, the lower-quartile contrast is larger than the contrasts at the median and upper quartile. Looking only at the average would obscure this pattern.

We study a common treatment date. Extending the framework to staggered adoption, where the set of untreated observations changes over time, is a natural next step. Other generative models may also be useful for questions about distributional shape.

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